

Practical No : 4

Aim: To implement multivariate Linear Regression

Code:

```
Import pandas as pd
Import seaborn as sns
Import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# 1. Select features and target
X = df[['Age', 'Pclass', 'SibSp', 'Parch']]
y = df['Fare']

print("First 5 rows of features (X):\n", X.head())
print("\nFirst 5 rows of target (y):\n", y.head())

# Handle missing values
X = X.dropna()
y = y.loc[X.index]

print("\nShape of data after removing NaN values:", X.shape)

# 2. Train-Test Split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
print("\nTraining set size:", X_train.shape)
print("Testing set size:", X_test.shape)

# 3. Train the model
model = LinearRegression()
model.fit(X_train, y_train)

print("\nModel Training Complete!")
print("Intercept (baseline Fare):", model.intercept_)
print("Coefficients (effect of each feature):")
for feature, coef in zip(X.columns, model.coef_):
    print(f" {feature}: {coef:.2f}")

# 4. Predictions
y_pred = model.predict(X_test)
print("\nFirst 10 Predictions:", y_pred[:10])
print("First 10 Actual Values:", y_test.values[:10])

# 5. Evaluation
print("\nMean Squared Error (MSE):", mean_squared_error(y_test, y_pred))
print("R2 Score (Explained Variance):", r2_score(y_test, y_pred))

# 6. Graphs for Better Understanding

# (a) Scatter plot: Actual vs Predicted
plt.figure(figsize=(6,6))
plt.scatter(y_test, y_pred, alpha=0.6, color="blue")
plt.xlabel("Actual Fare")
plt.ylabel("Predicted Fare")
plt.title("Actual vs Predicted Fare")
plt.plot([0, max(y_test)], [0, max(y_test)], 'r--') # ideal line
plt.show()
```

```
# (b) Residuals (errors)
residuals = y_test - y_pred
plt.figure(figsize=(6,4))
sns.histplot(residuals, bins=30, kde=True, color="purple")
plt.title("Distribution of Residuals (Errors)")
plt.xlabel("Error (Actual - Predicted)")
plt.show()

# (c) Feature importance bar chart
plt.figure(figsize=(6,4))
sns.barplot(x=X.columns, y=model.coef_, palette="coolwarm")
plt.title("Feature Importance (Coefficients)")
plt.ylabel("Effect on Fare")
plt.show()
```

[8]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...)	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

First 5 rows of features (X):

	Age	Pclass	SibSp	Parch
0	22	3	1	0
2	26	3	0	0
3	35	1	1	0
4	35	3	0	0
5	29	3	0	0

First 5 rows of target (y):

0	7.2500
2	7.9250
3	53.1000
4	8.0500
5	8.4583

Name: Fare, dtype: float64

Shape of data after removing NaN values: (769, 4)

Training set size: (615, 4)

Testing set size: (154, 4)

Model Training Complete!

Intercept (baseline Fare): 44.238234702161336

Coefficients (effect of each feature):

Age: 0.02
Pclass: -12.27
SibSp: 5.24
Parch: 4.40

First 10 Predictions: [8.01224152 16.2416578 36.03439102 7.97174339 20.24281544 20.06057382
8.19448314 8.05273966 20.16181917 36.95258073]

First 10 Actual Values: [7.7375 15.7417 21. 6.975 26. 13. 7.8958 7.8542 13.
55.]

Mean Squared Error (MSE): 67.0651440014923

R² Score (Explained Variance): 0.530410452586419

